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Introduction to Object Detection with CNNs

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Getting Started with Object Detection

Submitted

Getting Started with Object Detection

Review Learning Objectives

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Due Feb 18, 11:59 PM IST

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To Pass 75% or higher

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Several questions in this quiz will refer to the UsingObjectDetectionModels.mlx file. Make sure to work through the Live Script to answer the questions.

1. The goal of objection detection is to1 / 1 point

- ☒ find the location and class of one or more objects in the same image
- ☐ return multiple labels to describe the overall image
- ☐ find the location of possible objects
- ☒ Correct

Detection models both locate and identify objects in images

2. What is the backbone in an object detection model?1 / 1 point

- ☐ The anchor boxes that provide initial location estimates.
- ☒ A series of layers that extracts features from an input image, often the feature extraction part of a classification CNN.
- ☐ The output layer that produces the location and classification of an object.
- ☒ Correct

Object detection models usually use a classification network to extract features to identify potential objects.

3. True or False: all YOLO detection frameworks require anchor boxes.1 / 1 point

- ☐ True
- ☒ False
- ☒ Correct

The newer YOLOX models do not require anchor boxes.

4. What is the purpose of anchor boxes?1 / 1 point

- ☐ To increase the range of object sizes that can be identified
- ☒ To provide initial guesses for object locations and sizes.
- ☐ To limit a detector to find objects only of prescribed sizes and ignore other sizes.
- ☒ Correct

Anchor boxes provide initial guesses that are then refined by the network to predict a final bounding box.

5. True or False: the choice of backbone determines the detection framework.1 / 1 point

- ☐ True
- ☒ False
- ☒ Correct

The neck and head architecture determine the detection framework. Different backbones can be used with the same framework.

Refer to the UsingObjectDetectionModels.mlx file for the remaining questions.

6. In the UsingObjectDetectionModels.mlx Live Script, which models detect all six people using the default confidence thresholds? Select all that apply.1 / 1 point

- ☒ v4dark

☒ Correct

YOLO v4 with the Darknet backbone found all the people.
- ☒ yoloXsmall

☒ Correct

YOLOX with the small backbone found all the people.
- ☐ v4tiny
- ☒ yoloXtiny

☒ Correct

YOLOX with the tiny backbone found all the people.

7. Apply the YOLO v4 Darknet detector (v4dark) to the visionteam1.jpg image using a confidence threshold of 0.05. In addition to people, what object or objects are detected?1 / 1 point

- ☒ umbrellas
- ☐ suitcase
- ☐ refrigerator
- ☐ All of the above
- ☒ Correct

The detector mistakes the fake wood decoration for umbrellas.

8. What is the purpose of the "Threshold" Name-Value pair when using the detect function?1 / 1 point

- ☐ To set an upper bound for detection time. Only detections found under the "Threshold" time are returned.
- ☒ To set a minimum confidence score for detections to be returned. Detections below the threshold are discarded.
- ☐ To eliminate overlapping bounding boxes on a single object.
- ☒ Correct

Only detections with confidence scores greater than or equal to the "Threshold" value are returned.

9. Of the four models you used in the reading, which model had the slowest prediction speed?1 / 1 point

- ☐ v4tiny
- ☒ v4dark
- ☐ yoloXtiny
- ☐ yoloXsmall
- ☒ Correct

YOLO v4 with the Darknet-53 backbone uses three detection heads. It has the largest model size and the slowest prediction speed.

